Tony Nguyen

(509)-957-8378 · nguyen.cong@northeastern.edu · tonixsmm.github.io · linkedin.com/in/tonixsmm

Education

M.S. in Artificial Intelligence, *Machine Learning* — *Expected May 2028*
Northeastern University — Seattle, WA

B.A. in Computer Science, *Data Science & Economics* — *May 2026*
B.B.A. in Business Administration, *Finance* — Spokane, WA
Gonzaga University | President's Honor List | Data Science TA | *Summa Cum Laude* | GPA: 3.96

Research & Industry Experience

Academic Coordinator — September 2026 – Present
Global Experience Office - Northeastern University — *Seattle, WA*

- Leading requirements and engineering to replace a 20-year-old NUin class registration system, mapping workflows and undocumented workarounds across 8 stakeholder groups to define system requirements and integrations.

Cofounder, Business Lead — August 2026 – Present
kapsl — *Seattle, WA*

- Co-founded a **Rust-based** AI inference runtime that governs **GPU memory** dynamically across concurrent models, positioned against static-partitioning approaches (Triton, vLLM) for shared-GPU inference workloads.
- Designed a **benchmarking methodology** for the runtime with repeated runs, statistical validation, and hashed provenance records for every result. Stress-tested memory admission control under concurrent load and over-requests, with no crashes and no measurable impact on co-resident models.
- Led go-to-market strategy and investor positioning, pitching at investor networking events.

AI / Machine Learning Engineer – Research Assistant — January 2025 – May 2026
Institute for Informatics & Applied Technology – Gonzaga University — *Spokane, WA*

- Designed and built the **Wazuh MCP Server**, an open-source Python server (**30+ endpoints**) enabling LLMs to query and analyze security events from a Wazuh SIEM platform through structured tool calls.
- Designed a benchmarking study comparing **skill integration** across frontier models against a ground-truth attack chain, demonstrating that domain knowledge closes the performance gap between smaller and larger models.
- Benchmarked **three LLMs** on **60+ CVE reports** using a three-stage RAG pipeline with specialized agents for retrieval, analysis, and remediation recommendation, focusing on insightfulness and verifiability.
- Built and benchmarked ZagAI, an agentic RAG study assistant, comparing retrieval and chunking strategies with observability instrumentation, and interviewed 5 faculty and 7 students to shape the design.
- Developed and delivered a workshop on **RAG systems and LLM evaluation** to **30+ participants**, covering classical metrics, agent-based evaluation, and responsible AI practices.

Data Engineer Intern — June 2025 – September 2025
KPIM Vietnam — *Hanoi, Vietnam*

- Designed and maintained **10+ data pipelines** in SQL, Python, and Microsoft Fabric for two Big Data projects, with schema validation and quality checks that made querying **50% faster** than the client's system.
- Built reusable tooling and documentation for self-service data across the client's **six hotel properties**.

Data Engineer – Innovation Intern — June 2024 – August 2024
KONE Elevators & Escalators — *Singapore*

- Built a centralized **Customer 360** by integrating **20 years** of data from **five countries**; aligned models to enterprise architecture and recommended schema optimizations to improve consistency and join performance.
- Developed a **Tableau** dashboard using SAP, Salesforce, and Excel data; created models and calculations to monitor customer behavior, service KPIs, and revenue in real time.
- Built AWS S3 ETL pipelines with automated integrity checks and partitioning, cutting retrieval latency by **40%**.

Data Science & Machine Learning Intern — June 2023 – August 2023
TecAlliance Vietnam — *Ho Chi Minh City, Vietnam*

- Engineered Python algorithms to parse data from **100+ wiring-diagram SVGs**, creating structured datasets for computer vision models (YOLO) and advanced ML pipelines.
- Deployed a Neo4j graph database for schematic data, making queries **80%** faster than client's database.
- Led weekly global meetings of **15 team members** to unify data engineering efforts.

Project

Minecraft AI Companion | **Team Leader** | **Senior Capstone** — August 2025 – May 2026

- Built an open-source Minecraft mod with a **two-model architecture**, tool calling, and safety guardrails.

Skills

- Python, PyTorch, SQL, scikit-learn, Hugging Face, RAG, agentic systems, LLM evaluation, MCP, LangChain